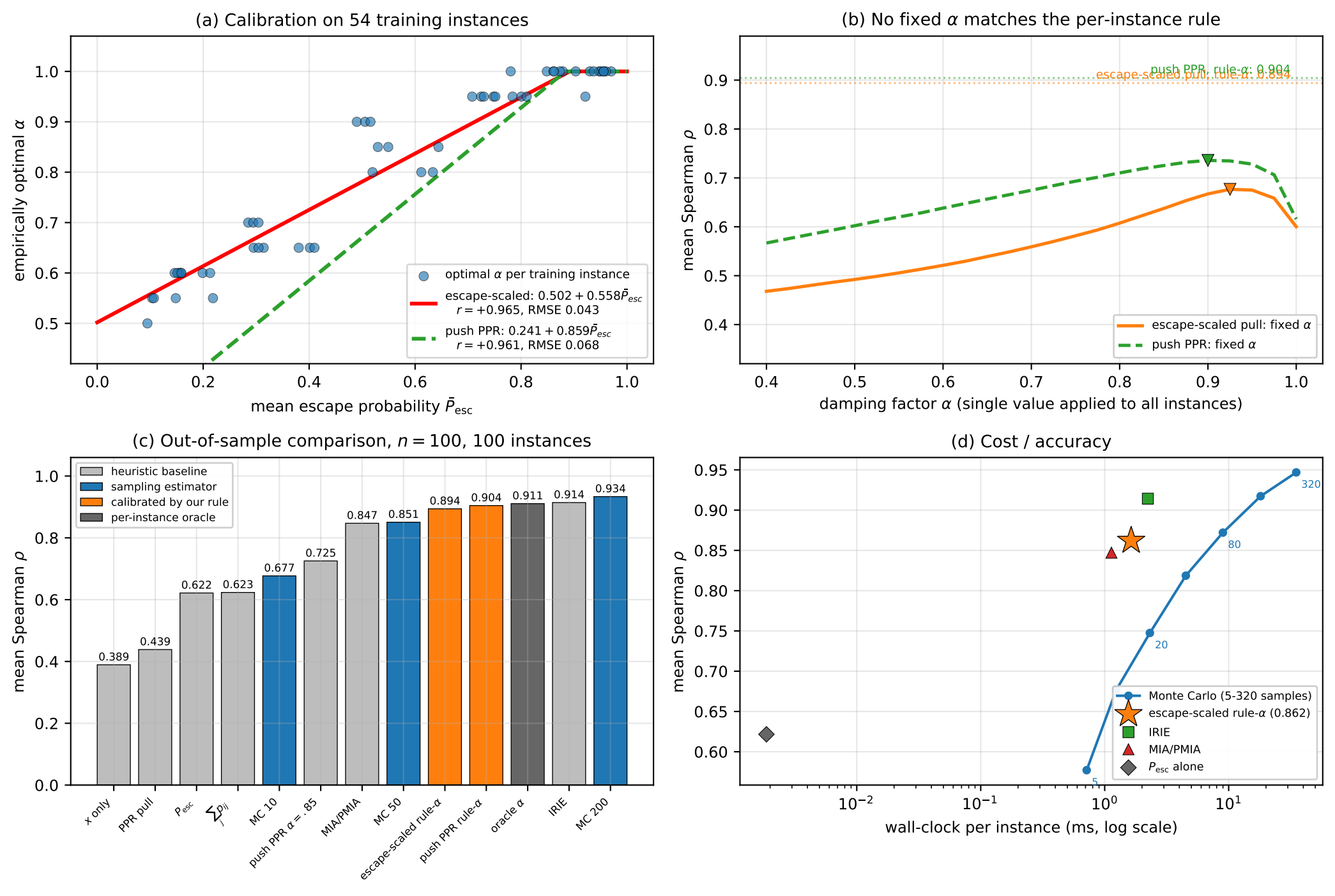
(a) Calibration on 54 training instances
empirically optimal α
mean escape probability $\bar{P}_{esc}$
optimal α per training instance
escape-scaled: $0.502 + 0.558\bar{P}_{esc}$
r = +0.965, RMSE 0.043
push PPR: $0.241 + 0.859\bar{P}_{esc}$
r = +0.961, RMSE 0.068
(b) No fixed α matches the per-instance rule
mean Spearman ρ
damping factor α (single value applied to all instances)
push PPR, rule-α: 0.904
escape-scaled pull, rule-α: 0.894
escape-scaled pull: fixed α
push PPR: fixed α
(c) Out-of-sample comparison, n = 100, 100 instances
mean Spearman ρ
heuristic baseline
sampling estimator
calibrated by our rule
per-instance oracle
0.389
0.439
0.622
0.623
0.677
0.725
0.847
0.851
0.894
0.904
0.911
0.914
0.934
x only
PPR pull
P_{esc}
$\sum_j p_{ij}$
MC 10
push PPR α = .85
MIA/PMIA
MC 50
escape-scaled rule-α
push PPR rule-α
oracle α
IRIE
MC 200
(d) Cost / accuracy
mean Spearman ρ
wall-clock per instance (ms, log scale)
Monte Carlo (5-320 samples)
escape-scaled rule-α (0.862)
IRIE
MIA/PMIA
P_{esc} alone
5
20
80
320